Module 09 – Fixed Charge Problem

Exploratory Data Analysis

Model Formulation

Min = y1+y2+y3+y4
Where WH = 1,2,3,4 and DC = 1,2,3,4,5,6
Binary = 0,1
DC1 = x1+x2+x3+x4 = 740
DC2 = x12+x22+x32+x42 = 888
DC3 = x13+x23+33+43 = 696
D4 = x14+x24+x34+x44 = 673
D5 = x15+x25+x35+x45 = 862
DC6 = x16+x26+x36+x46 = 516
WH 1,2,3,4 >=0

Model Optimized for Min Costs to Supply DCs

										Set up Costs	
WH v DC >	1	2	3	4	5	6	WH Sum Sent	Binary Viaribles	Linking Constraints	Actual	Possible
1	0	0	0	673	0	0	0	0	0	0	2874
2	0	0	0	0	0	0	0	0	0	0	2680
3	0	0	0	0	862	516	1378	1	-2997	2601	2601
4	740	888	696	0	0	0	2324	1	-2051	2490	2490
Shipped	740	888	696	673	862	516					
Required	740	888	696	673	862	516	4375				
Total cost	$ 68,966										

My model represents the optimal way to supply all distribution centers using only the necessary warehouses, minimizing total costs. This is done by fully meeting the demand at each distribution center, selecting only the cost effective warehouses of 3 & 4, and minimizing the total cost by balancing shipping and set up costs.

Model with Stipulation

1. *Instead of only being able to open 2 warehouses, what happens to our objective function when we only can open 1 warehouse?*

										Set up Costs	
WH v DC >	1	2	3	4	5	6	WH Sum Sent	Binary Viaribles	Linking Constraints	Actual	Possible
1	0	0	0	673	0	0	0	0	0	0	2874
2	0	0	0	0	0	0	0	0	0	0	2680
3	0	0	0	0	0	0	0	0	0	0	2601
4	740	888	696	0	862	516	3702	1	-673	2490	2490
Shipped	740	888	696	673	862	516					
Required	740	888	696	673	862	516	4375				
Total cost	$ 83,142										

When only one warehouse is open, the total cost increases because its limits routing flexibility, often leading to higher shipping expenses that outweigh the savings from reduced setup costs.

1. *For distance between each location, we used Manhattan distance but what happens to our model if we use Euclidean distance instead? Did the change impact the model at all? Do you feel this is a better distance metric to use in this scenario?*

										Set up Costs	
WH v DC >	1	2	3	4	5	6	WH Sum Sent	Binary Viaribles	Linking Constraints	Actual	Possible
1	0	0	0	673	0	0	0	0	0	0	2874
2	0	0	0	0	0	0	0	0	0	0	2680
3	0	0	0	0	0	0	0	0	0	0	2601
4	740	888	696	0	862	516	3702	1	-673	2490	2490
Shipped	740	888	696	673	862	516					
Required	740	888	696	673	862	516	4375				
Total cost	$ 20,650										

When using the Euclidean distance instead of the Manhattan distance, it only changes the total cost, leading to a decrease which is good in this scenario and it would be better to use the Euclidean distance for cheaper distribution from warehouse to the distribution centers.